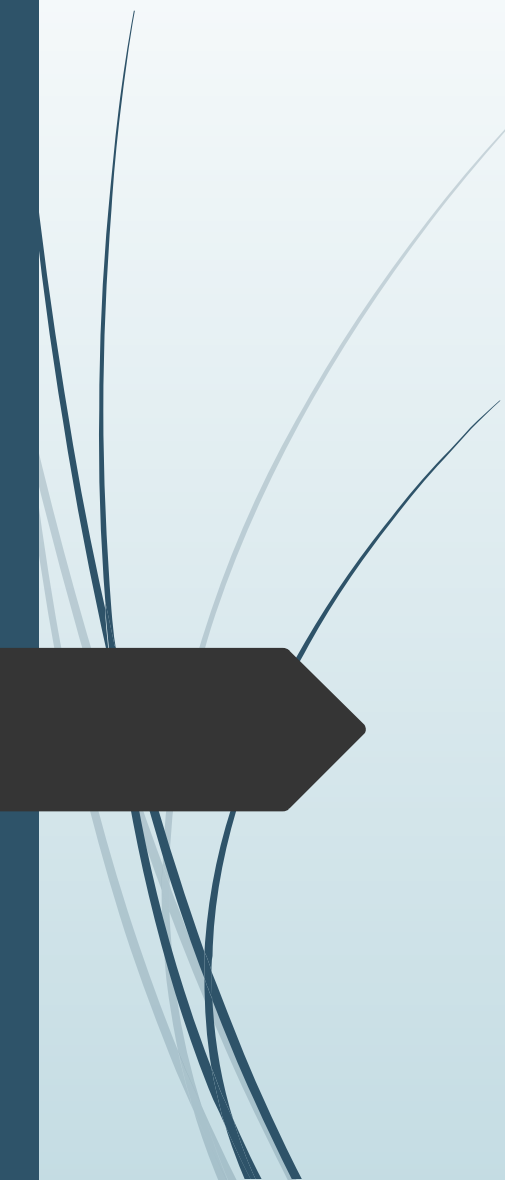


Introduction to Artificial Intelligence

Week 9

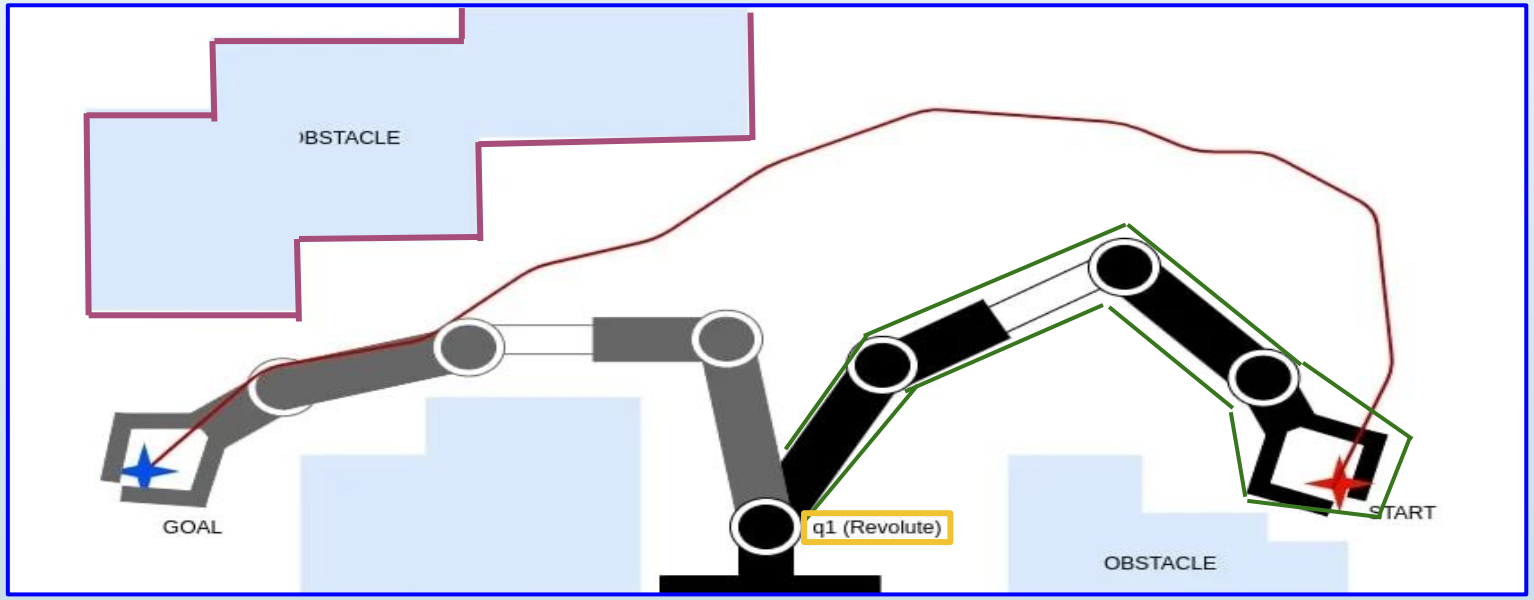
Planning Domain Definition Language (PDDL)

Introduction

- As planning has evolved, so has the language used to describe it
- Planning Domain Definition Language (PDDL) is a family of languages which allow us to define a planning problem
- PDDL one of the few languages designed for the purpose of creating a standard for Artificial Intelligence (AI) planning

Components of PDDL

- A **world** in PDDL is described by set of states each containing a list of facts and/or objects
- A world begins with an **initial state**
- A world governed by a set of rules and **constraints** that limit which actions can be taken in each state
- Each **action** generally represents a **transition** to a different state



Components of PDDL

- There are certain things we need to keep track of in the "world":
  - **Objects**: Things in the world that interest us
  - **Predicates**: Facts that we are interested in (e.g. properties of objects), which can be true or false
  - **Actions/Operators**: Ways of changing the state of the world, i.e. things that happen that change the facts
  - **An initial state**: The state of the world that we start in, i.e. things that are true at the start
  - **Goal specification**: The state of the world we want to end at, i.e. things that we want to be true at the end

PDDL Syntax

- First thing you need to know, PDDL files usually have the extension **.pddl**

- There are two PDDL files you need to learn the syntax of:
  - The Domain File **(domain.pddl)**
    - establishes the context of the world
    - determines what sorts of details the states can include (predicates)
    - what can we do to move between states in the world (actions).

  - The Problem File **(problem.pddl)**
    - represents an instance of the world we established in the domain
    - determines what is true at the start of the plan (initial state)
    - what we want to be true at the end of the plan (goal state).

PDDL Syntax

- **The Domain File structure (domain.pddl)**

```
(define (domain <domain name>)

        (:predicates
                <predicate-list>
        )

         ; optional block
        (:types
                <types-list>
        )

        (:action
                <action-details>
        )
)
```

PDDL Syntax

- **The Problem File structure (problem.pddl)**

```
(define (problem <title>)

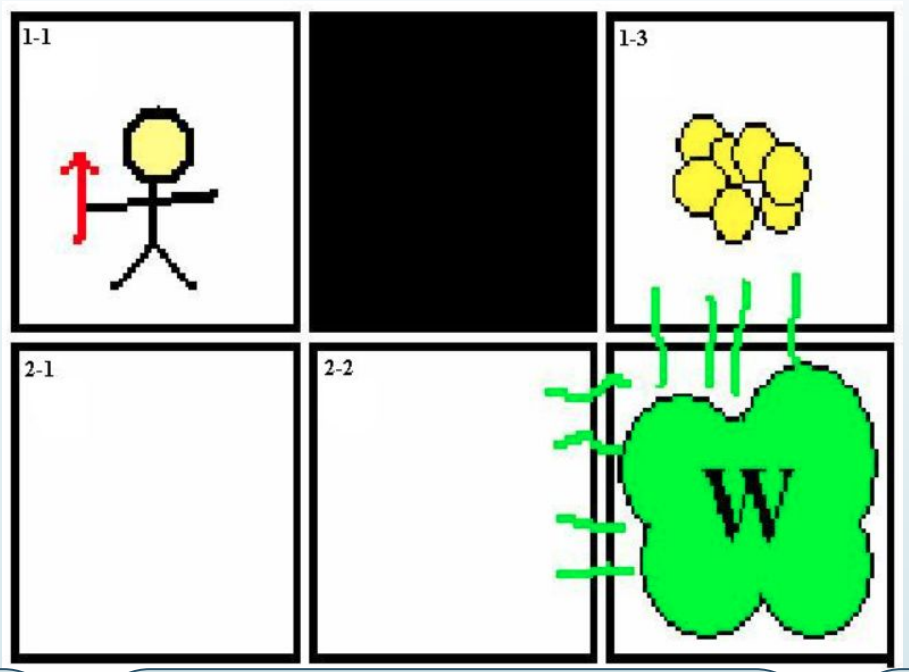
        (:domain <domain-name>)

        (:objects
                <object-list>
        )

        (:init
                <predicates>
        )

        (:goal
                <predicates>
        )
)
```

Example (WUMPUS WORLD)



Environment

- sq 1-1 is agent
- sq 1-2 is pit
- sq 1-3 is gold
- sq 2-3 contain Wumpus

Actions

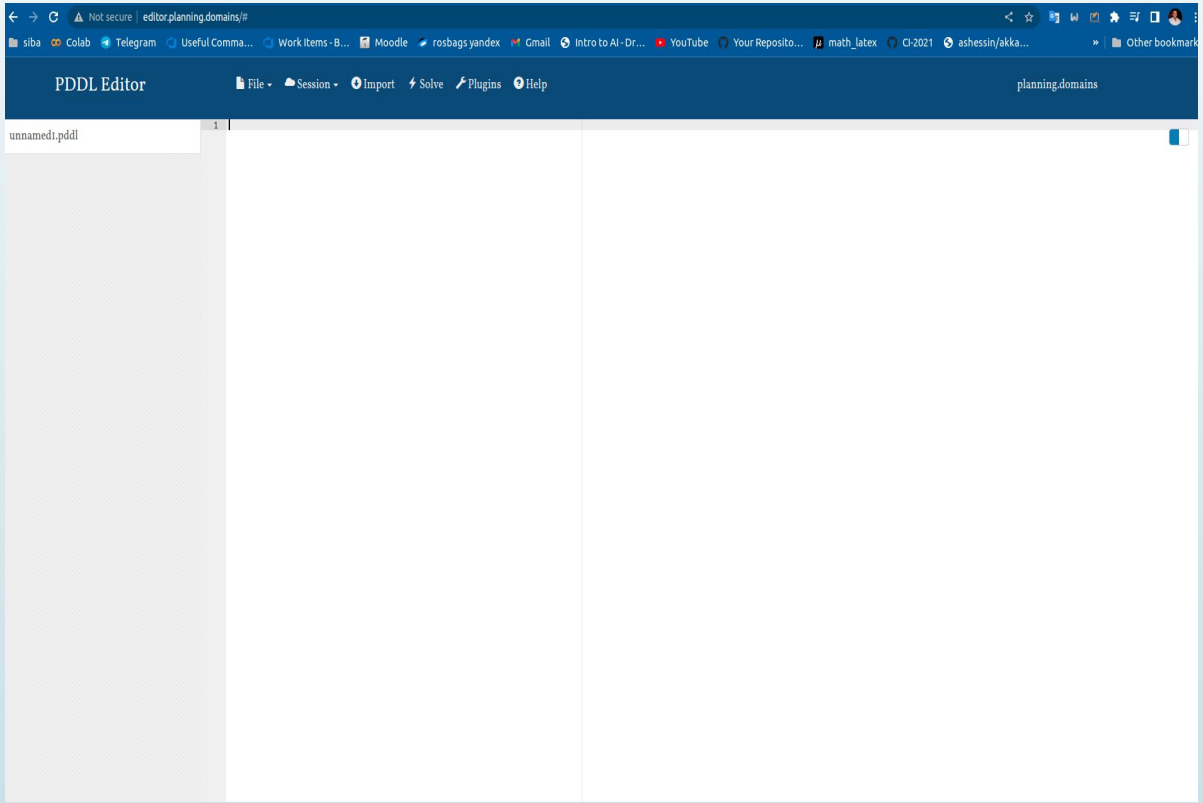
- Move from sq to adj sq
- Kill Wumpus with arrow
- Take Gold

Goal

- Get gold & Return back to start pos

Environment Setup

- We will use [PDDL Online Editor]



SCAN ME